

Problem Set 5

*Each student must submit your own **HANDWRITTEN** work. Turn in a paper copy at the beginning of lecture on 11/2/2026. Please write your name on each page and make sure your problem set is stapled. Students are allowed to work in groups, but each student must turn in their own set of answers. No help from outside of the course is allowed. To receive full credit for a problem, all work must be shown. Partial credit will be awarded. All problems are weighted evenly.*

1. Suppose that a firm's short-run total cost function is given by $STC(Q) = Q^2 + 200Q + 100$

(a) Write the firm's short-run profit maximization problem

(b) Solve for the firm's short-run supply function.

(c) For what range of prices will the firm choose not to produce any quantity of output?

Now suppose that this is the only firm in the market and that market demand is given by $Q_D = 950 - 2P$

(d) Solve for the short-run market equilibrium quantity and price.

2. Suppose that each firm in a perfectly competitive market has long-run cost represented as $TC(Q) = 100Q^2 - 100Q + 100$.

(a) What are the three conditions for a long-run perfectly competitive equilibrium?

(b) If the firm's goal is to maximize profit in a perfectly competitive market, find the quantity that each firm will choose to sell.

(c) At what price will firms sell each unit?

(d) Suppose that the market demand is given by $Q_D = 2000 - 5P$. How many firms will enter the market in the long run?

Now suppose that, due to changing tastes, consumer demand has shifted outward, such that $Q_D = 2400 - 6P$

- (e) Find the new price and quantity that each firm will sell the good at in the long-run (assuming constant input prices).

- (f) How many additional firms will enter the market, in response to the increased demand?

3. Suppose that domestic demand is given by $Q_D = 5000 - 100P$ and the domestic supply curve is given by $Q_S = 150P$. The world price for this good is $P = 10$. Suppose that a 50% tariff on the price of this good is being considered.
- (a) Draw a graph illustrating the free trade equilibrium (with no tariff). Label all relevant parts of the graph including equilibrium price and quantity & producer and consumer surplus.
- (b) By how much would the tariff increase producer surplus for domestic suppliers?
- (c) How much would the government collect in tariff revenues?
- (d) What is the dead-weight loss from the tariff?